

UNIVERSAL WEB SCRAPER

Chrome Extension Development Plan

Professional Development Proposal

May 2026

Executive Summary

This document outlines the complete development plan for a general-purpose web scraper Chrome extension. The extension enables users to visually select data fields from any website and scrape information with multi-page support, automatic pagination handling, and flexible data export options.

Key Features

- Point-and-click field selection on any website (Shopify, WooCommerce, Amazon, Facebook, Instagram, and more)
- Smart pagination detection and automatic multi-page scraping
- Support for multiple data types (text, links, images, prices)
- Flexible export to CSV, JSON, and Excel formats
- Save and reuse scraping templates for repeated tasks

1. Technology Stack

Architecture

- **Manifest V3** (Latest Chrome Extension standard)
- **Content Script:** Injected into websites for DOM inspection
- **Service Worker:** Background processing for pagination loops
- **Chrome Storage API:** Local data persistence and templates

Languages & Frameworks

- Vanilla JavaScript (ES6+) — lightweight, no heavy dependencies

- HTML5 & CSS3 — clean, semantic markup
- Optional: Tailwind CSS for polished UI

2. Development Phases

Phase 1: Foundation & Setup

Deliverables

- Project scaffolding and folder structure
- manifest.json configuration (Manifest V3)
- Basic popup UI with initial controls
- Extension loads successfully in Chrome

Phase 2: Field Selection System

Deliverables

- Content script for hovering and element inspection
- Visual highlighting of hoverable elements
- XPath and CSS selector generation
- Selected fields display and management UI

Phase 3: Data Extraction Engine

Deliverables

- Core scraper engine for data extraction
- Support for text, links, images, and prices
- Row and column layout handling
- Single-page scraping with progress feedback

Phase 4: Pagination & Multi-Page Support

Deliverables

- Automatic "Next" button detection
- Smart page load detection
- Multi-page data aggregation
- Pause/resume/stop controls

Phase 5: UI/UX Polish

Deliverables

- Progress bars and status indicators
- Data preview functionality
- Professional interface design
- Tooltips and help text

Phase 6: Data Export

Deliverables

- JSON export with full metadata
- CSV export with proper escaping
- Excel (.xlsx) export with formatting
- One-click download functionality

Phase 7: Templates & Configuration

Deliverables

- Options page with settings
- Save/load scraping templates
- Auto-detect and suggest templates
- Template management (edit, delete, export)

Phase 8: Advanced Features

Deliverables

- Field transformations (regex, trim, case conversion)
- Filtering and deduplication
- Data preview and editing
- Comprehensive testing and bug fixes

3. Core Features & Capabilities

Feature 1: Point-and-Click Field Selection with Live XPath Display

Users can hover over website elements to highlight them, then click to select fields for scraping. Works on any website regardless of how it is built — Shopify, WooCommerce, Amazon, Facebook, Instagram, or any custom JavaScript framework. As the user moves their cursor over any element on the page, the extension instantly displays that element's XPath in a tooltip or side panel — in real time. This helps advanced users verify exactly what they are selecting before clicking.

- Visual feedback with element highlighting
- Hover over any element — XPath is shown instantly in real time
- XPath displayed in a live tooltip or side panel as cursor moves
- XPath is copyable — user can copy it for use elsewhere
- Field naming and type assignment

Feature 2: Smart Data Extraction

The extension extracts multiple data types from each element, automatically handling different content types and formats.

- Text content extraction
- Link URLs and titles
- Image sources (including lazy-loaded images)
- Price parsing and normalization
- Custom attributes and metadata

Feature 3: Automatic Pagination & Unlimited Scrolling

The extension handles both paginated sites (Next button) and infinite scroll sites (like Instagram, Facebook, LinkedIn). Scrolling continues automatically and without limit until all data is loaded or the user stops it.

- Auto-detect Next buttons (10+ patterns supported)
- Smart page load detection
- Duplicate detection to prevent re-scraping
- Unlimited scrolling — scrapes until all content is loaded
- Works on infinite scroll sites — Instagram, Facebook, LinkedIn, etc.
- Configurable delay between scroll steps and page loads

Feature 4: Flexible Data Export

Scraped data can be exported in multiple formats for analysis in spreadsheets, databases, or custom applications.

- JSON — Complete data with metadata
- CSV — Delimited text format
- Excel (.xlsx) — Formatted spreadsheets with headers
- Column selection and custom naming

Feature 5: Template System

Save scraping configurations as templates for websites you scrape frequently. Templates are automatically suggested when you revisit sites.

- Save templates by URL pattern
- Auto-load templates on recognized sites
- Template management (edit, delete, export)
- Usage analytics per template

Feature 6: AI Auto-Detection

When the extension is opened on any page, it automatically analyzes the HTML structure using AI heuristics to detect and highlight the most relevant data — tables, product listings, directories, and more. No manual selection required to get started.

- Auto-detects tables, lists, and repeating elements
- AI predicts the most relevant data on the page
- User can accept auto-detection or manually adjust
- Works on any website without site-specific configuration

Feature 7: Live Data Preview

Before exporting, users see a live table preview of all scraped data inside the extension popup. This allows them to verify the data is correct and make adjustments before downloading.

- Real-time table preview inside popup
- Shows all rows and columns as they are scraped
- Updates live during multi-page scraping
- Copy and paste support from the preview table

Feature 8: Column Renaming

Users can rename any column in the preview table before exporting. This makes exported data clean and ready to use without needing to edit the spreadsheet afterwards.

- Click any column header to rename it
- Rename technical names to friendly labels (e.g. span.price to Product Price)
- Renamed columns saved in templates for reuse
- Exported file uses the renamed column headers

Feature 9: Column Delete & Filtering

Users can remove unwanted columns from the preview table before exporting. Only the columns they need are included in the final downloaded file.

- Delete any unwanted column with one click
- Filter rows by column value before export
- Remove empty or duplicate rows
- Final export only contains selected columns and filtered rows

4. Testing & Quality Assurance

Testing Websites

The extension will be tested on diverse websites to ensure robust compatibility:

- E-commerce sites (Amazon, eBay, Shopify, WooCommerce)
- Social media (Facebook, Instagram listings)
- News and content sites
- HTML tables and data-heavy pages
- Complex nested layouts

Quality Metrics

- Scrape 100+ items without data loss
- Support for 5+ page pagination
- Data integrity across all export formats
- No crashes or critical errors in normal use

5. Technical Details

External Dependencies

Minimal dependencies approach: Core functionality uses only vanilla JavaScript

- **Optional:** `xlsx.js` (~125KB) for Excel export
- **Optional:** Tailwind CSS for enhanced UI

Security & Privacy

- All processing happens locally — no data sent to servers
- No tracking or analytics
- Rate-limiting to avoid server overload
- Clear user consent for each action
- User responsible for respecting website ToS

Data Storage

- Chrome Storage API for templates and settings
- ~10MB local storage for datasets
- Optional: IndexedDB for very large datasets

6. Use Cases

Business Applications

- Price monitoring across multiple e-commerce sites
- Competitor product listings and reviews
- Real estate and property listings
- Job listings and salary research

Research & Analytics

- Market research and trend analysis
- Content aggregation for multiple sources
- Academic data collection
- Compliance monitoring

7. Technical Challenges & Solutions

Challenge	Solution
Dynamic Content (JS-rendered)	Use MutationObserver to detect DOM changes; wait for content before extracting
Lazy-Loaded Images	Scroll elements into view before extracting src/data-src attributes
Fragile Selectors	Provide multiple selector options; allow manual refinement
Page Detection	Monitor URL + DOM changes + validate new content
Large Datasets	Stream to file; use chunked processing
Performance	Use Service Worker for background processing; batch operations
CORS	Handled naturally by browser; extract from already-loaded DOM

8. Future Roadmap

- Cloud sync of templates across devices
- Community template marketplace
- Scheduled/recurring scraping
- Database integration (direct SQL export)
- Mobile companion app
- API for programmatic scraping
- AI-powered field detection

9. Comparison: Our Extension vs Instant Data Scraper

The table below compares our custom-built extension against Instant Data Scraper, one of the most popular free scraping tools available. Our solution matches all core features while adding exclusive capabilities not available in any free tool.

Feature	Instant Data Scraper	Our Extension
AI Auto-Detection	✓ Yes	✓ Yes
Point & Click Selection	✓ Yes	✓ Yes
Works on Any Website	✓ Yes	✓ Yes
Dynamic JS Content	✓ Yes	✓ Yes
Unlimited Scrolling	✓ Yes	✓ Yes

Pagination / Next Button	✓ Yes	✓ Yes
Live XPath on Hover	✗ No	✓ Yes — Exclusive
Live Data Preview	✓ Yes	✓ Yes
Column Renaming	✓ Yes	✓ Yes
Column Delete & Filter	✓ Yes	✓ Yes
Export to CSV & Excel	✓ Yes	✓ Yes
Delay / Speed Settings	✓ Yes	✓ Yes
Save & Reuse Templates	✗ No	✓ Yes — Exclusive
Auto-Load by Website	✗ No	✓ Yes — Exclusive
Custom Branding	✗ No	✓ Yes — Exclusive
Full Ownership	✗ No	✓ Yes — You own it
Cost	Free (generic)	One-time investment

Key Advantage: Instant Data Scraper is a generic free tool with no template system, no branding, and no customization. Our extension is built specifically for your workflow — with saved templates, auto-load on recognized sites, and full ownership of the tool.

10. Next Steps

1. Client review and approval of this plan
2. Design UI/UX mockups for popup and scraping interface
3. Setup development environment and repository
4. Begin Phase 1 development
5. Regular progress updates and milestone reviews